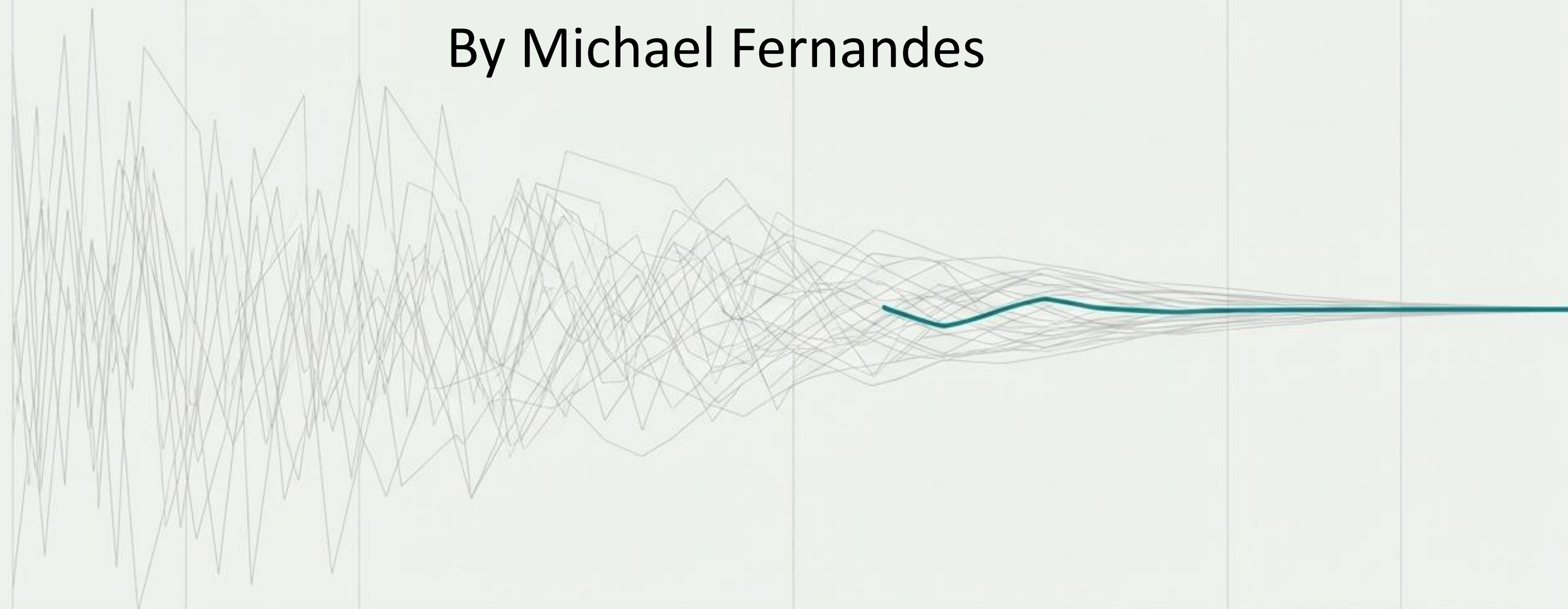


Time Series in Quant

From Discrete-Time Stochastic Processes to Alpha Generation

By Michael Fernandes

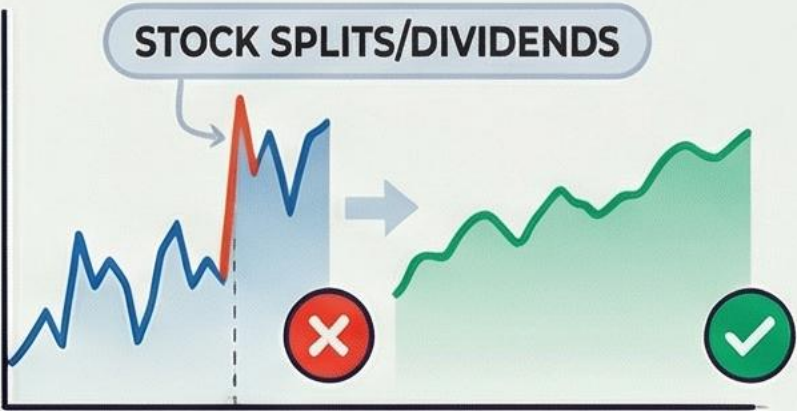


The Quant's Playbook: From Raw Data to Market Signals

Outline of the quantitative financial time series analysis workflow, from data preparation to modeling and risk management.

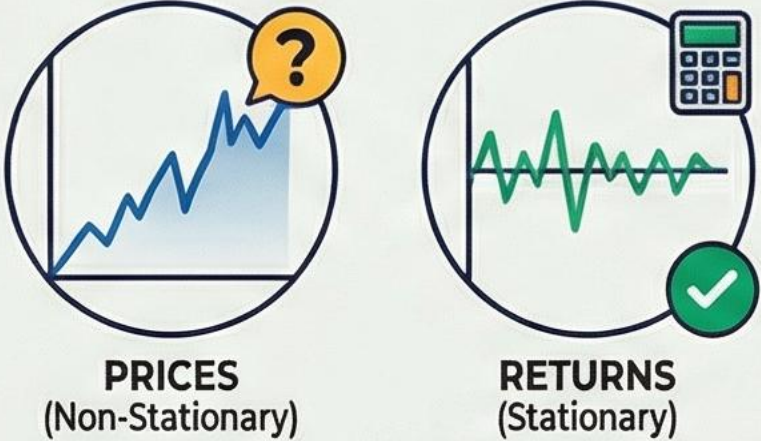
STAGE 1: BUILDING A SOLID DATA FOUNDATION

Raw Prices Lie: You Must Adjust for Corporate Actions



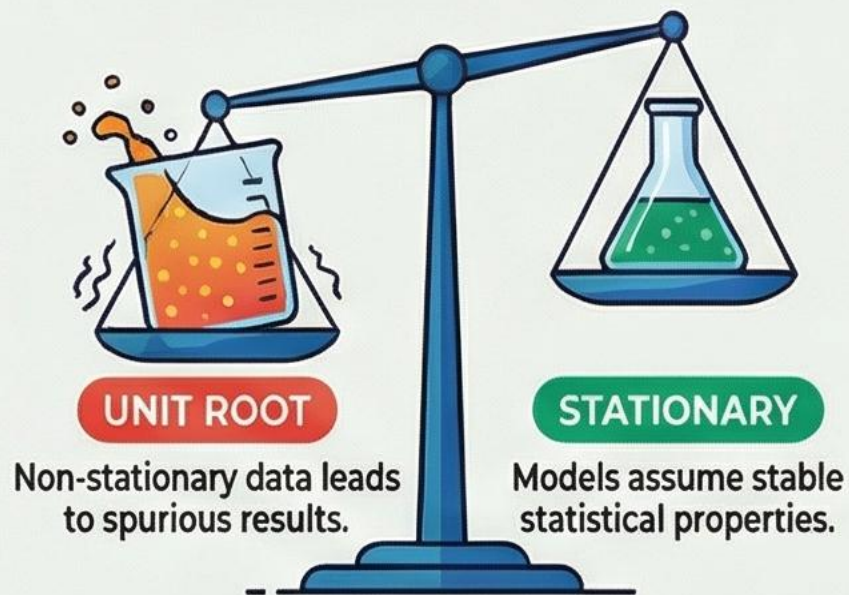
Stock splits and dividends create artificial price jumps that distort volatility and returns.

Price vs. Returns: Choosing the Right Transformation



Prices are non-stationary; models analyze returns. Use Simple Returns for portfolios, Log Returns for time series.

The Quest for Stationarity: Is Your Data Stable?



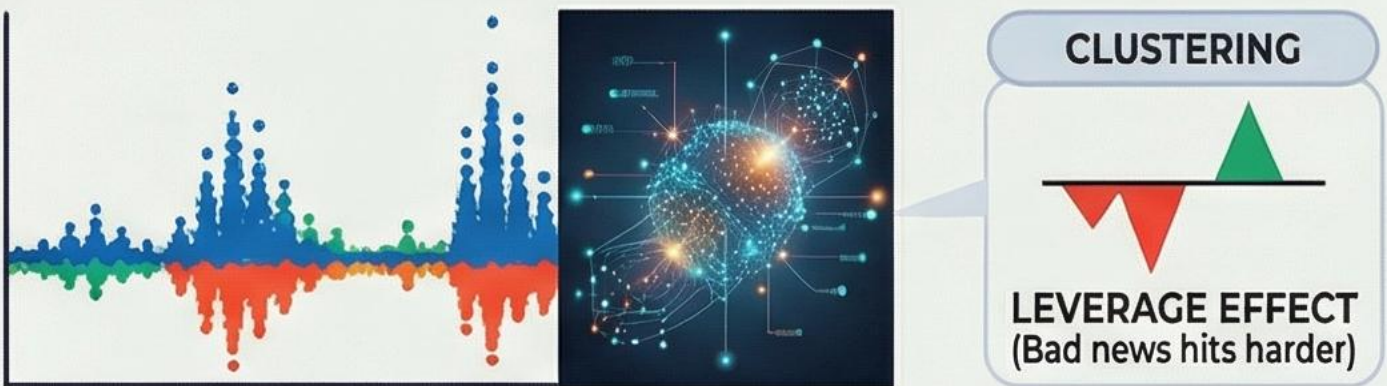
Non-stationary data leads to spurious results.

Models assume stable statistical properties.

STATIONARITY TEST INTERPRETATION (ADF & KPSS)		
	ADF Test Result (H_0 : Unit Root)	KPSS Test Result (H_0 : Stationary)
Reject H_0	Fail to Reject H_0	STATIONARY (Safe for modeling)
Fail to Reject H_0	Reject H_0	UNIT ROOT (Needs differencing)

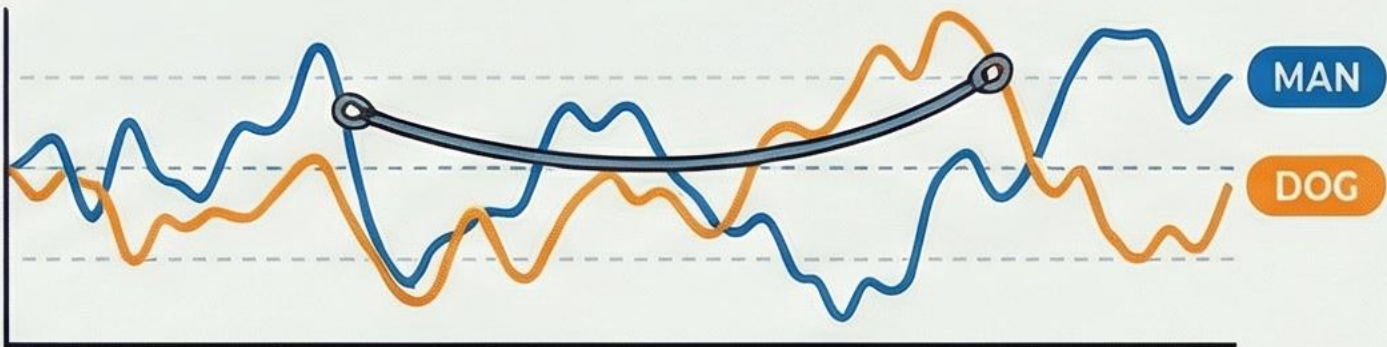
STAGE 2: MODELING SIGNALS & AVOIDING TRAPS

Modeling Volatility: The GARCH Framework



Volatility isn't constant. GARCH models capture "volatility clustering" and the "leverage effect".

Finding Equilibrium: Cointegration & The "Drunk Man's Dog"



Two assets may wander randomly (like a man and his dog) but are linked by a long-term relationship (a leash).

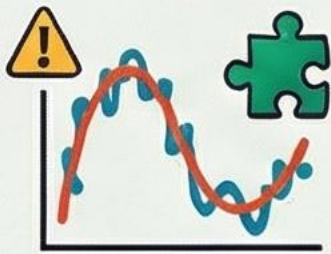
The Three Deadly Sins of Quantitative Research



LOOK-AHEAD BIAS
Using future information in past simulations.



SURVIVORSHIP BIAS
Ignoring failed assets that no longer exist.



OVERFITTING
Creating a model too complex that fails on new data.

Beware! These create FALSE PROFITS.

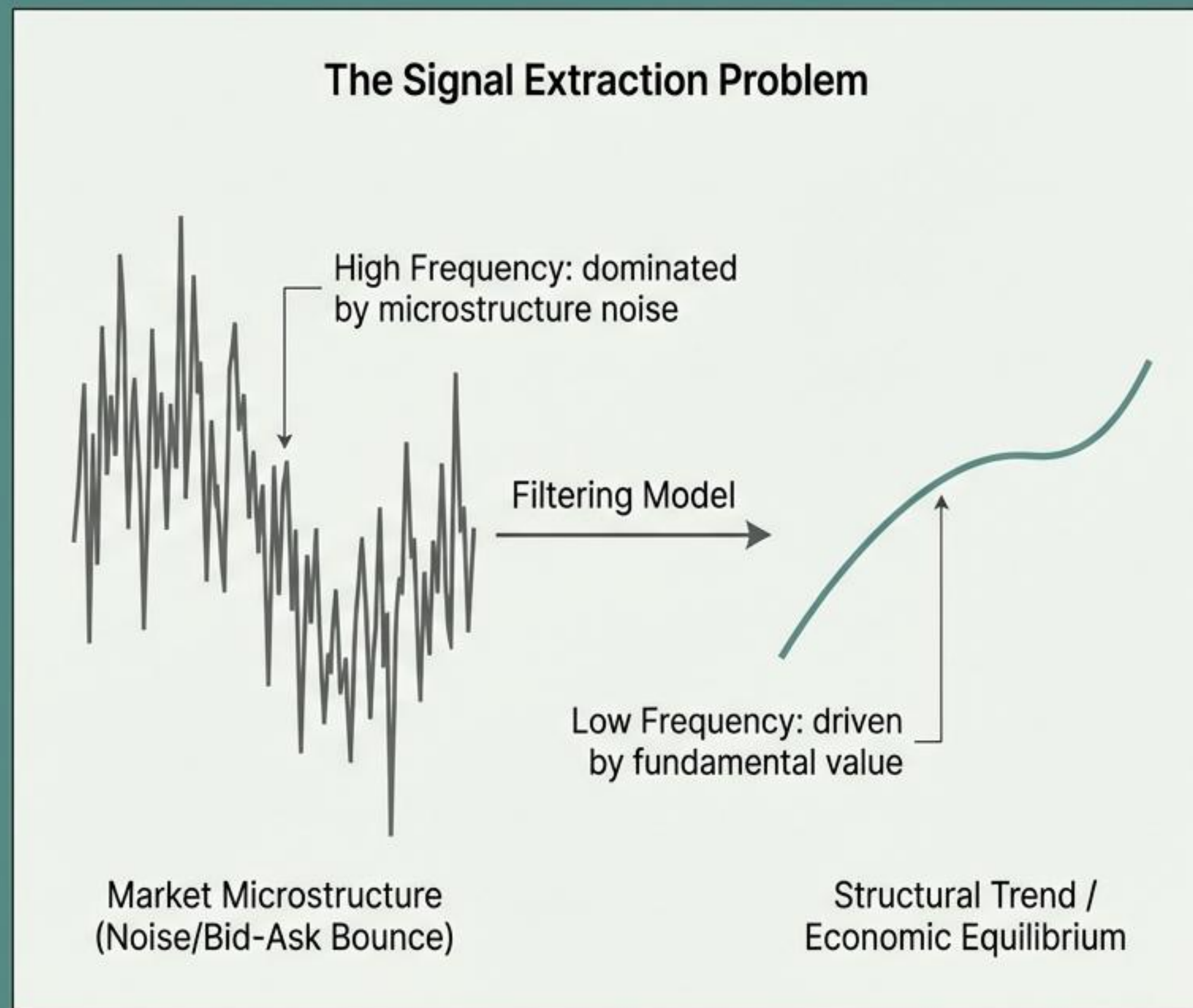
Financial Data is a Discrete-Time Stochastic Process

****Core Philosophy****

We do not observe deterministic numbers. We observe a single “sample path” of a hidden generating mechanism.

****The Objective****

Inference: Uncovering the probability distributions governing transitions ($P_t \rightarrow P_{t+1}$) to predict future values or risk.



Data Engineering I: The Architecture of Price

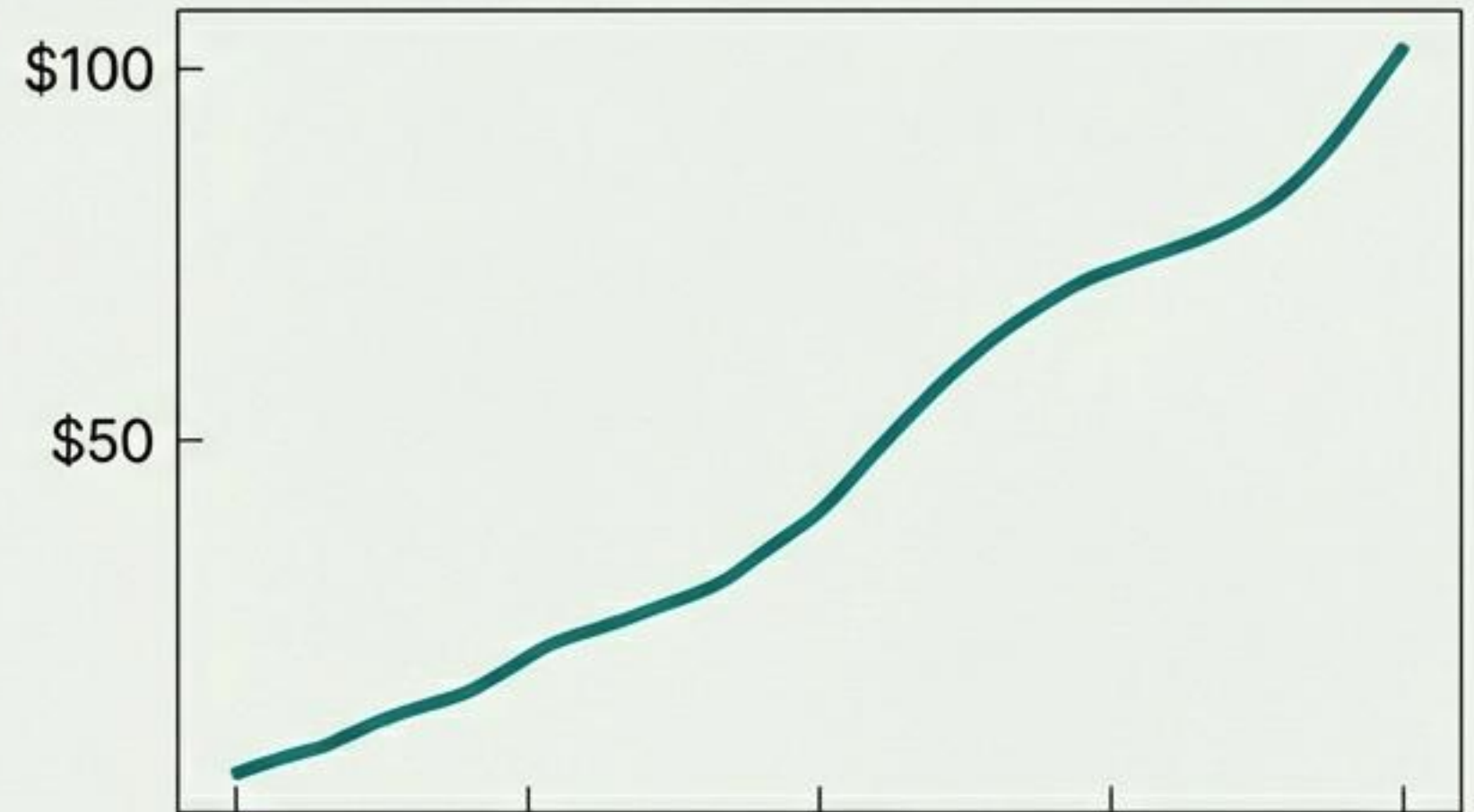
The impact of Corporate Actions on Algorithmic Interpretation

Raw Price Series (Unadjusted)



Algorithm Interpretation: 50% Crash.
Triggers false stop-loss. Volatility σ
Volatility σ estimate spikes to infinity.

Adjusted Price Series



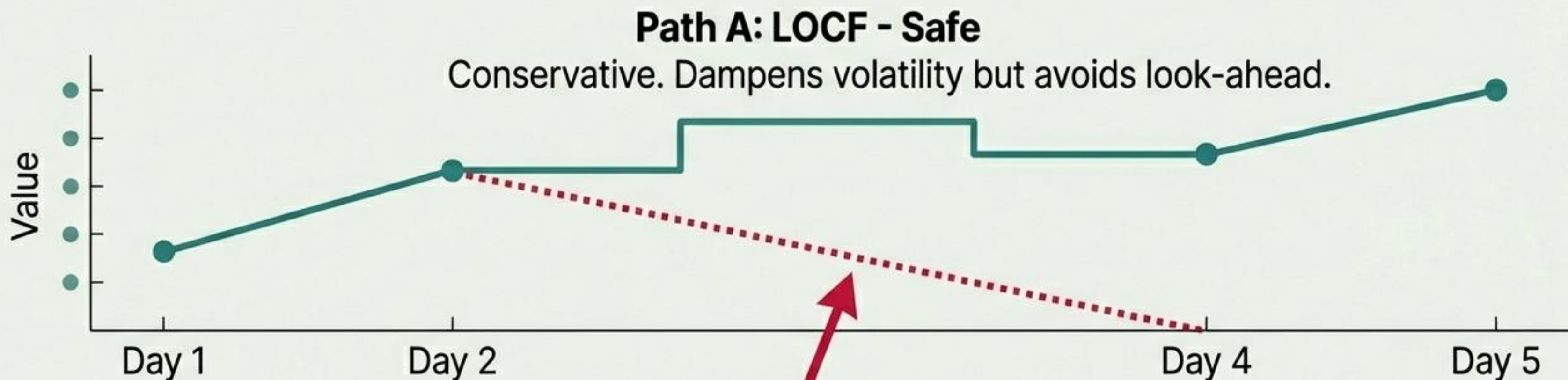
Correct Interpretation: Value Neutral.
Smooth continuity preserves statistical
validity.

Data Engineering II: Dividend Adjustments

	Backward Adjustment 	Forward Adjustment 
Mechanism	Adjusts historical prices downward based on payouts.	Fixes start price, adjusts future upward (reinvestment).
Primary Use Case	Technical Analysis / Visual Charting.	Institutional Performance (TSR) & Wealth Compounding.
The Trade-off	Pro: Matches current ticker price. Con: Distorts history (prices appear near zero in distant past).	Pro: Mathematically accurate for compounding. Con: Diverges from current market price.

Data Engineering III: The Danger of Holes

Imputation Bias: How we fill missing data determines model validity.



Uses the future (Day 4) to calculate the past (Day 3).
Leaks future information into the training set.

The Mathematics of Returns

Simple Returns (R_t)

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Key Property:

Cross-sectionally Additive (Linear across assets).

Use Case:

Portfolio Optimization ($\sum w_i R_i$).

Log Returns (r_t)

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Key Property:

Time Additive (Linear across time).

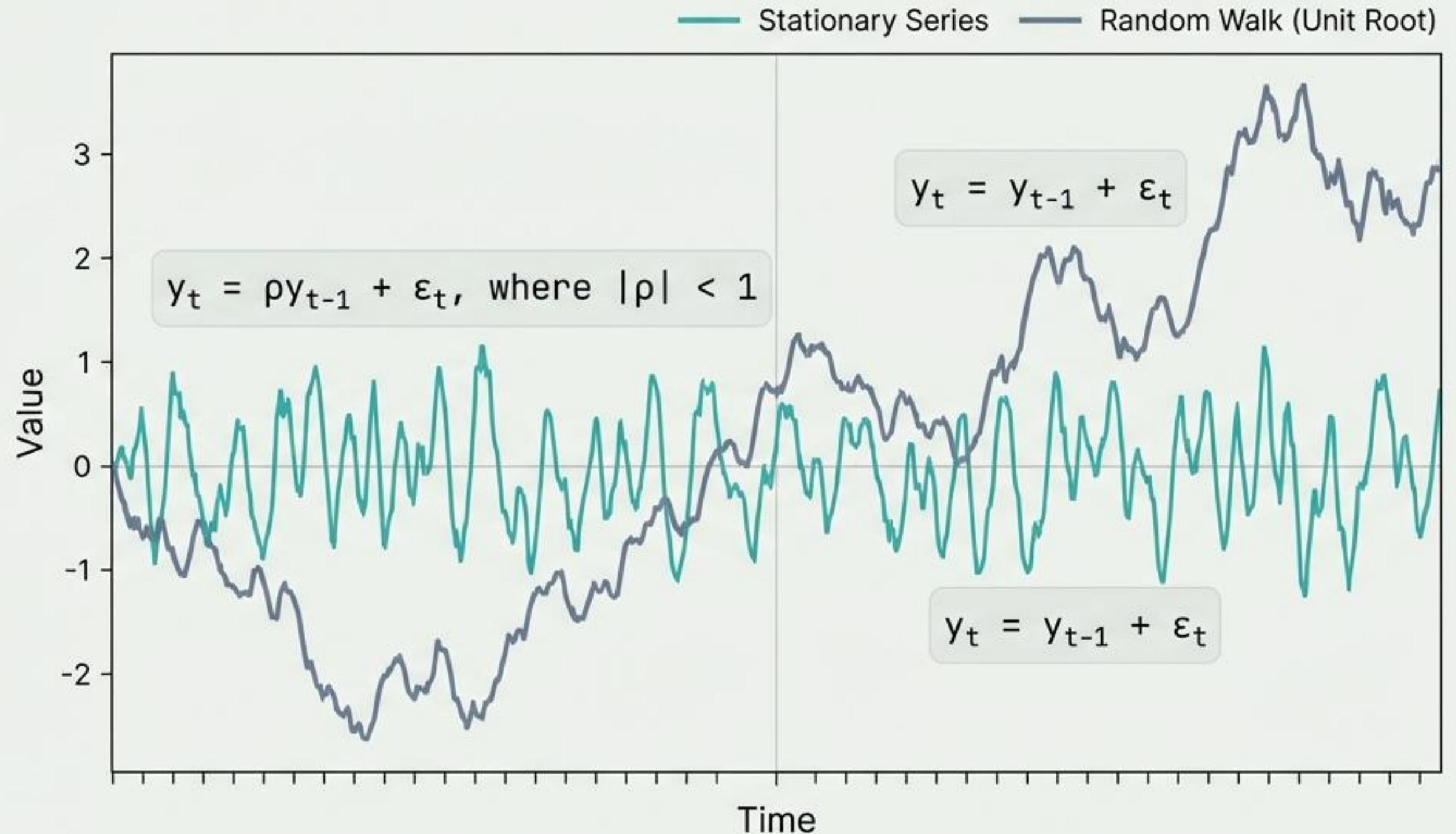
Use Case:

Volatility Modeling & Risk ($\sum r_{t+i}$).

Note: $\ln(1+x) \approx x$ for small intervals, but divergence creates wealth errors over long horizons.

Stationarity and the Unit Root Problem

A model requires a stable target. If mean and variance change over time (Unit Root), the model fails.



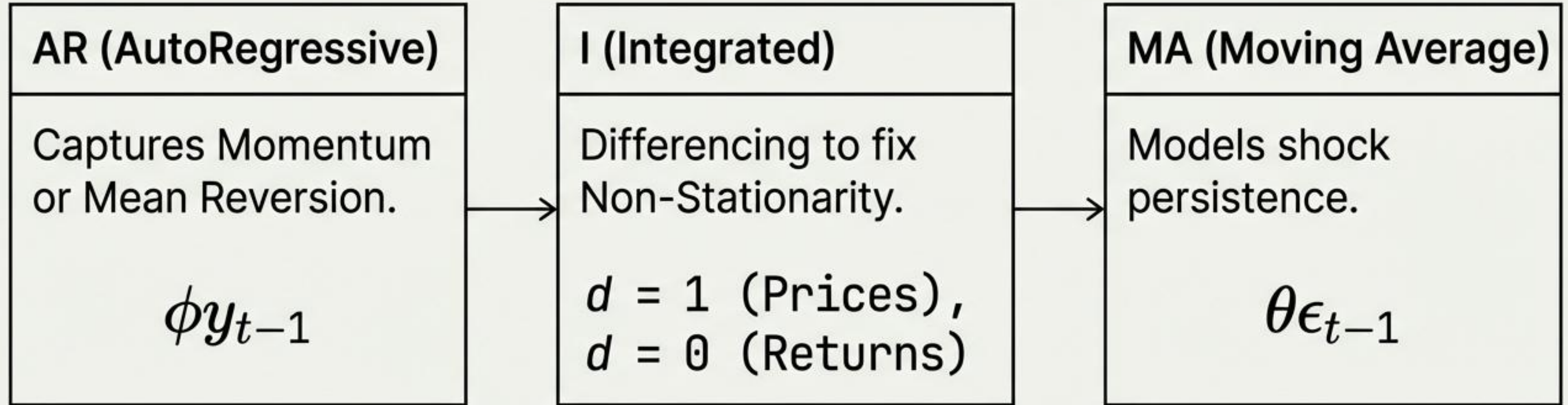
Triangulating the Truth: Stationarity Tests

Using conflicting tests to categorize time series behavior.

		KPSS Test (H_0 : Stationary)	
		Fail to Reject	Reject
ADF Test (H_0 : Unit Root)	Reject	**STATIONARY** Safe for modeling.	
	Fail Reject	**TREND STATIONARY** Requires Detrending.	**UNIT ROOT** Random Walk. Needs Differencing.
		Fail to Reject	Reject
		KPSS Test (H_0 : Stationary)	

Linear Stochastic Models (ARIMA)

AutoRegressive Integrated Moving Average

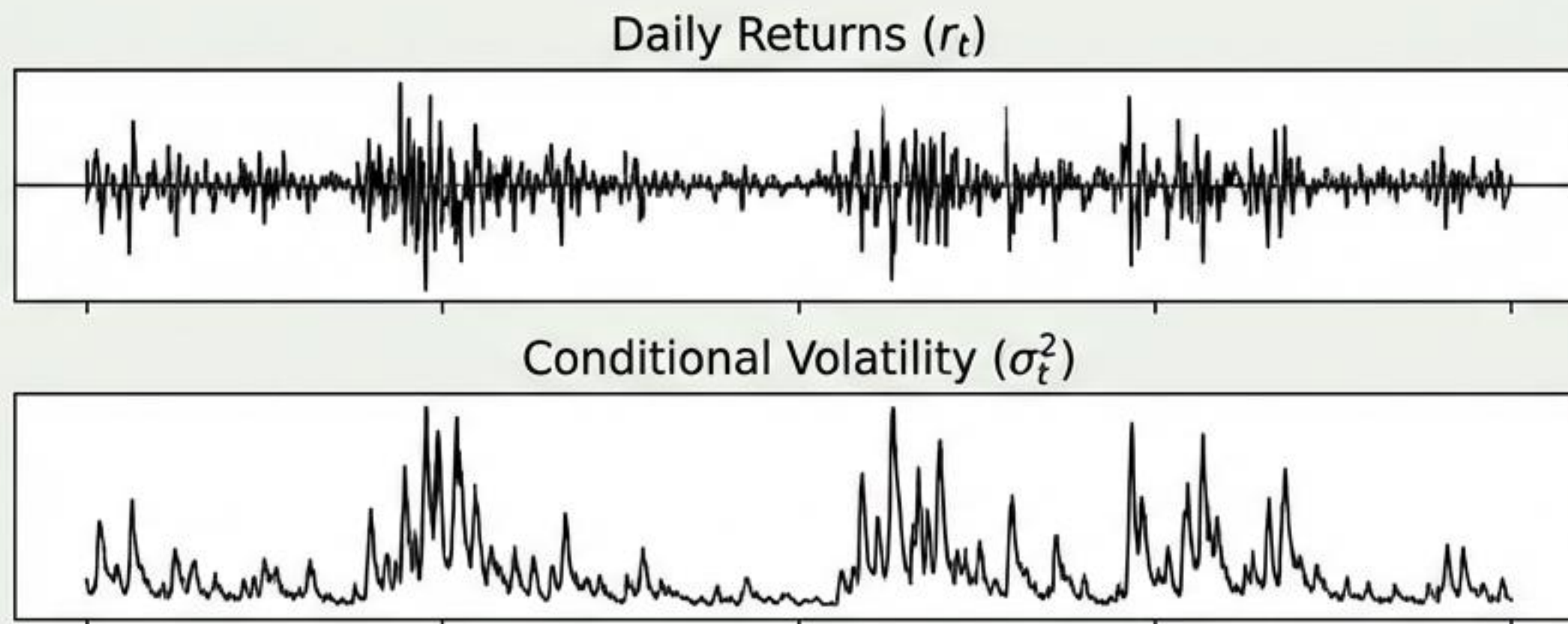


The Limitation

Good for macroeconomics (inertial trends). Often fails for asset returns (Efficient Markets implies Random Walk). Returns are random, but **risk** is not.

Modeling Volatility Clustering (GARCH)

Stylized Fact: Large moves follow large moves. Variance is autocorrelated.



$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

The "Jumpiness"
(Reaction to new shock)



The "Persistence"
(Memory of fear)



The Leverage Effect & Asymmetry (EGARCH)

The Flaw in GARCH:

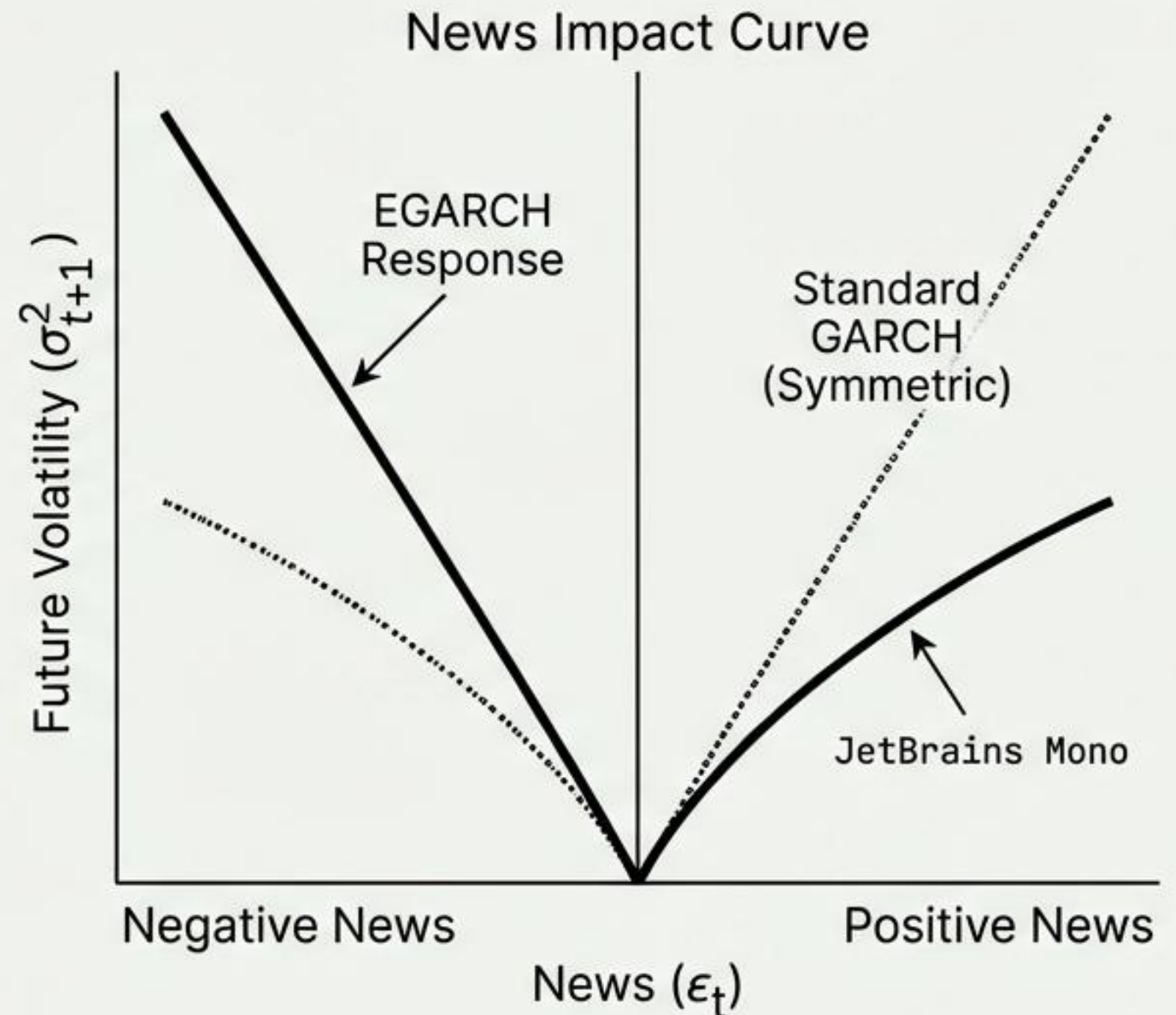
It squares the shock (ϵ^2), treating panic and exuberance identically.

The Reality:

Markets take the stairs up and the elevator down.

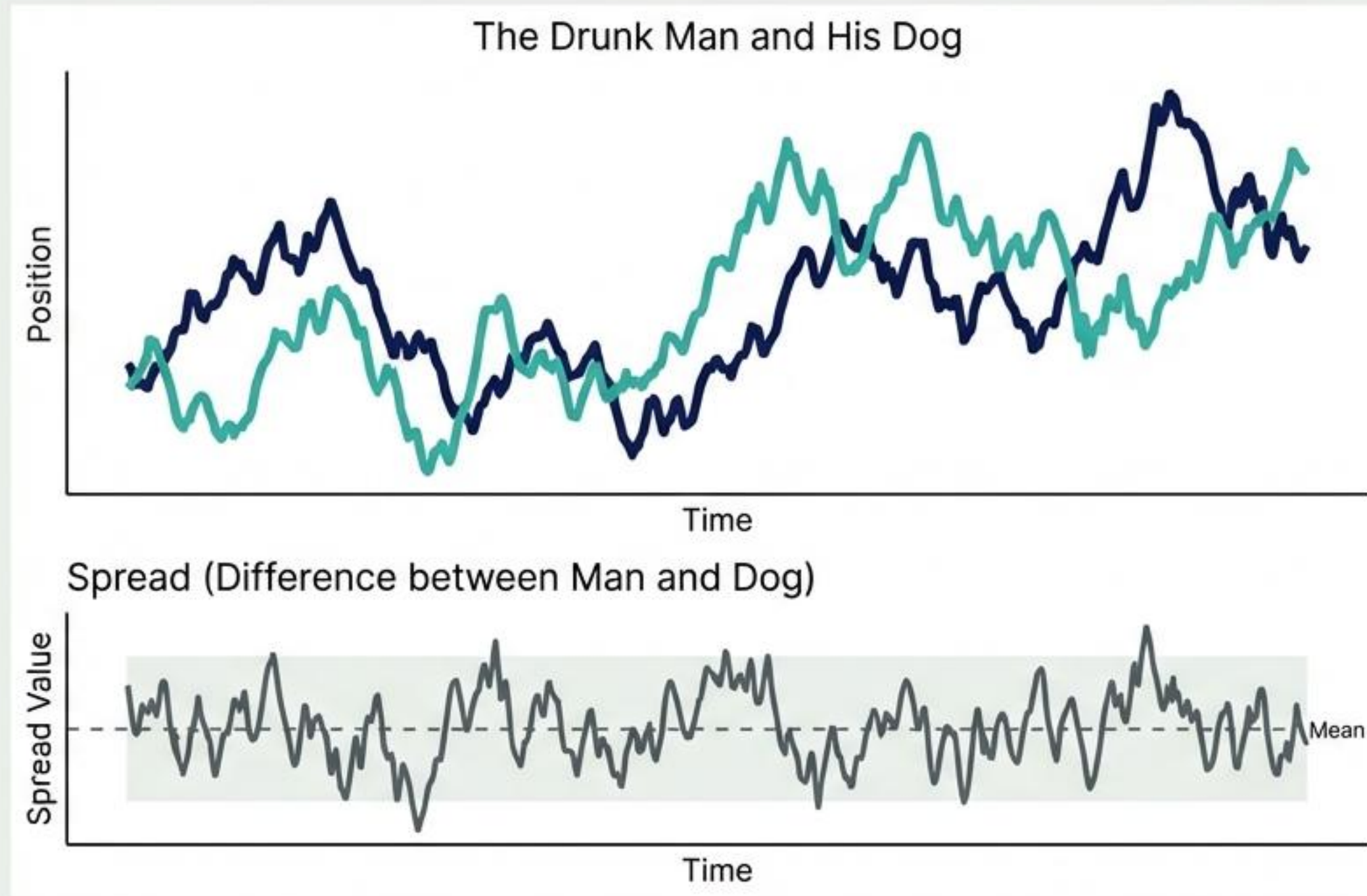
Mechanism:

- Price Drop $\rightarrow$ Higher Debt/Equity Ratio $\rightarrow$ Higher Risk.



Multivariate Dynamics: Correlation vs. Cointegration

Analogy Visual: The Drunk Man and His Dog.



Correlation:

Short-term sync. Are they stepping left together?

Cointegration:

Long-term tether. Is there a leash?

Trading Alpha:

We bet on the leash pulling them back together.

Detection Methodologies: The Nuclear Option

Engle-Granger Method

scope: Pair Trading (Y vs X).



Process: Run OLS $\rightarrow$ Test Residuals (ADF).

Limitation: Dependent on direction (Y on $X \neq X$ on Y).

Johansen Test

scope: Portfolios / Baskets.



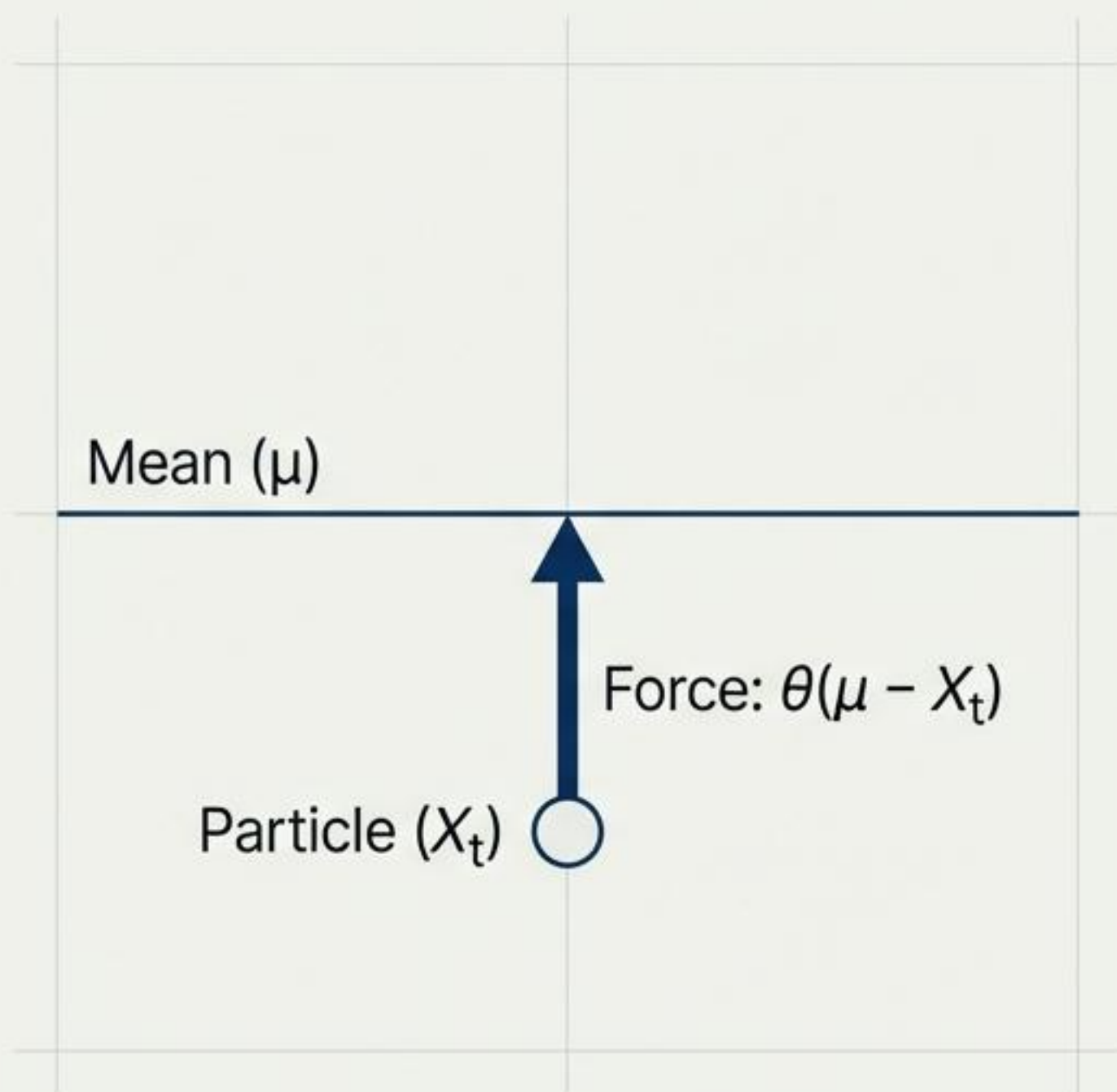
Process: Vector Error Correction Model (VECM). Solves Eigenvalue problem.

Advantage: Symmetric. Finds complex relationships (e.g., 1 Google = 0.5 Apple + 0.3 Microsoft).

Johansen is the superior tool for hedged portfolios.

Mean Reversion Strategy (Ornstein-Uhlenbeck)

Calibrating the 'Spring' of the market.



$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

Parameter List & The Quant Trick

θ (Theta): Speed of Reversion.

High θ = Quick Profit.

Low θ = High Carrying Risk.

The Quant Trick:

Regress X_{t+1} on X_t to get ϕ .

Derive $\theta = -\ln(\phi)/\Delta t$.

Result: Precise Z-score entry calculation.

Trend Following & Time Series Momentum (TSMOM)

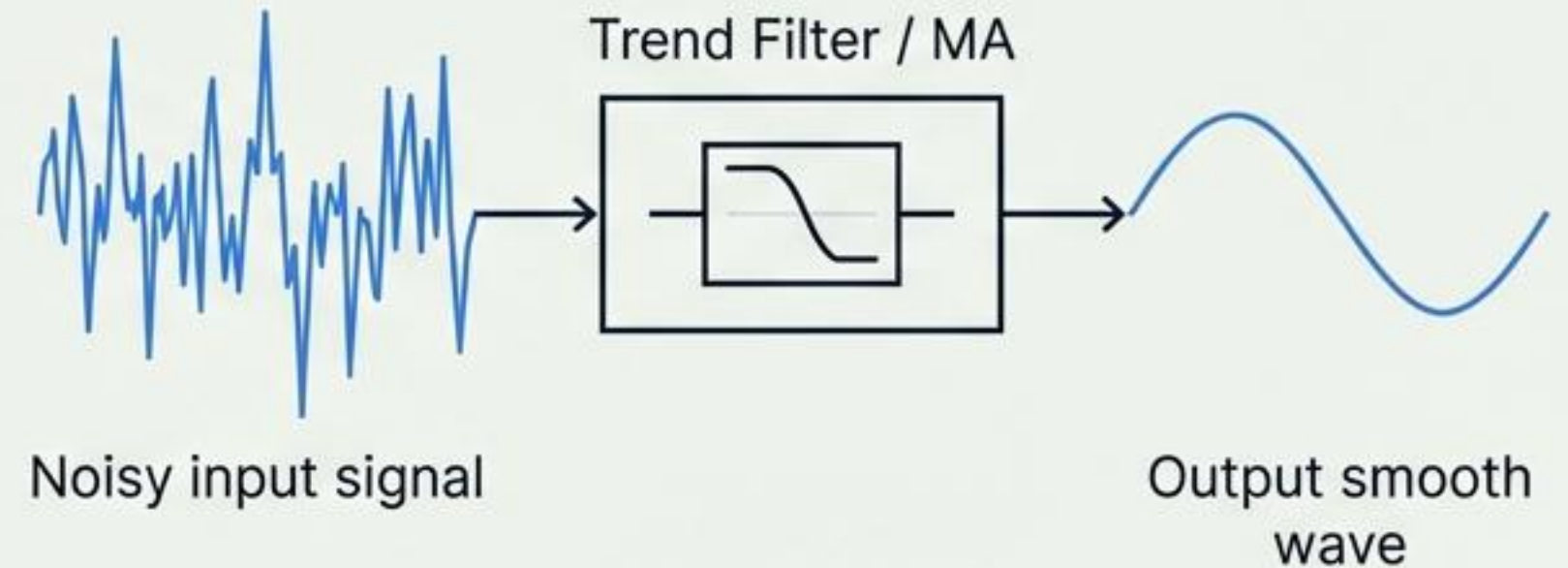
While Mean Reversion bets on the rubber band snapping back, Momentum bets on the trend continuing.

Crisis Alpha (2008)



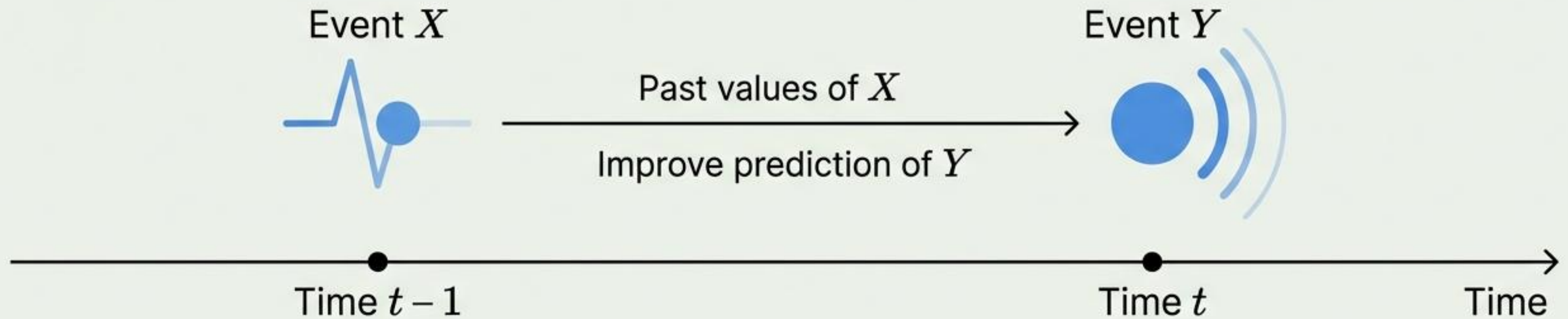
TSMOM acts as a crisis hedge. In 2008, trends were strong (Bonds Up, Commodities Down).

Low Pass Filter Methodology



Granger Causality & Predictive Precedence

X Granger-causes Y if past values of X improve prediction of Y .



Warning: Predictive Precedence $\neq$ Physical Causality.

The rooster crows before sunrise, but does not cause the sun to rise.

Ljung-Box Test:

Applied to residuals.
If $p < 0.05$, signal
remains on the table.

The Seven Sins I: Look-Ahead Bias

Using data you shouldn't have.

Common Pitfalls

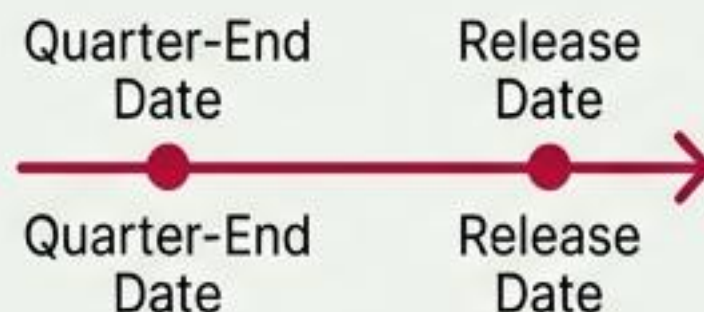
1



The High/Low Trap

Trading on daily Highs/Lows.
Impossible to know until
market close.

2



The Earnings Drift

Using earnings data on the
quarter-end date vs. the
actual release date.

3



Imputation Errors

Linear interpolation of
missing values (leaking
future).

The Seven Sins II: Survivorship & Overfitting

Survivorship Bias

S&P 500 company



Microsoft

amazon

Google



LEHMAN BROTHERS

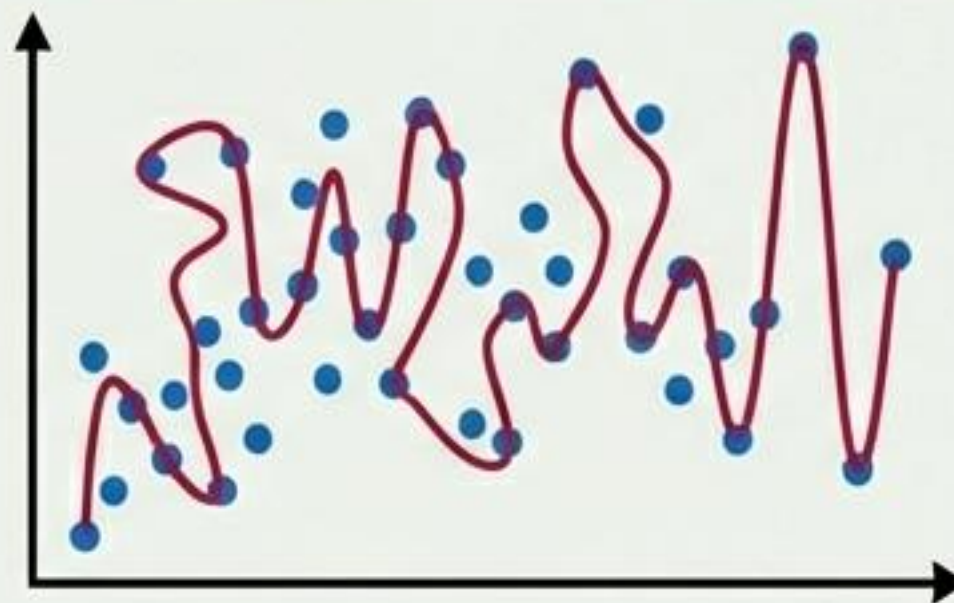
WORLD COM

Sears

Testing only on current winners inflates Sharpe Ratios. Historical data must include delisted and bankrupt entities for accurate backtesting.

Warning: Excluding failures creates a false illusion of success.

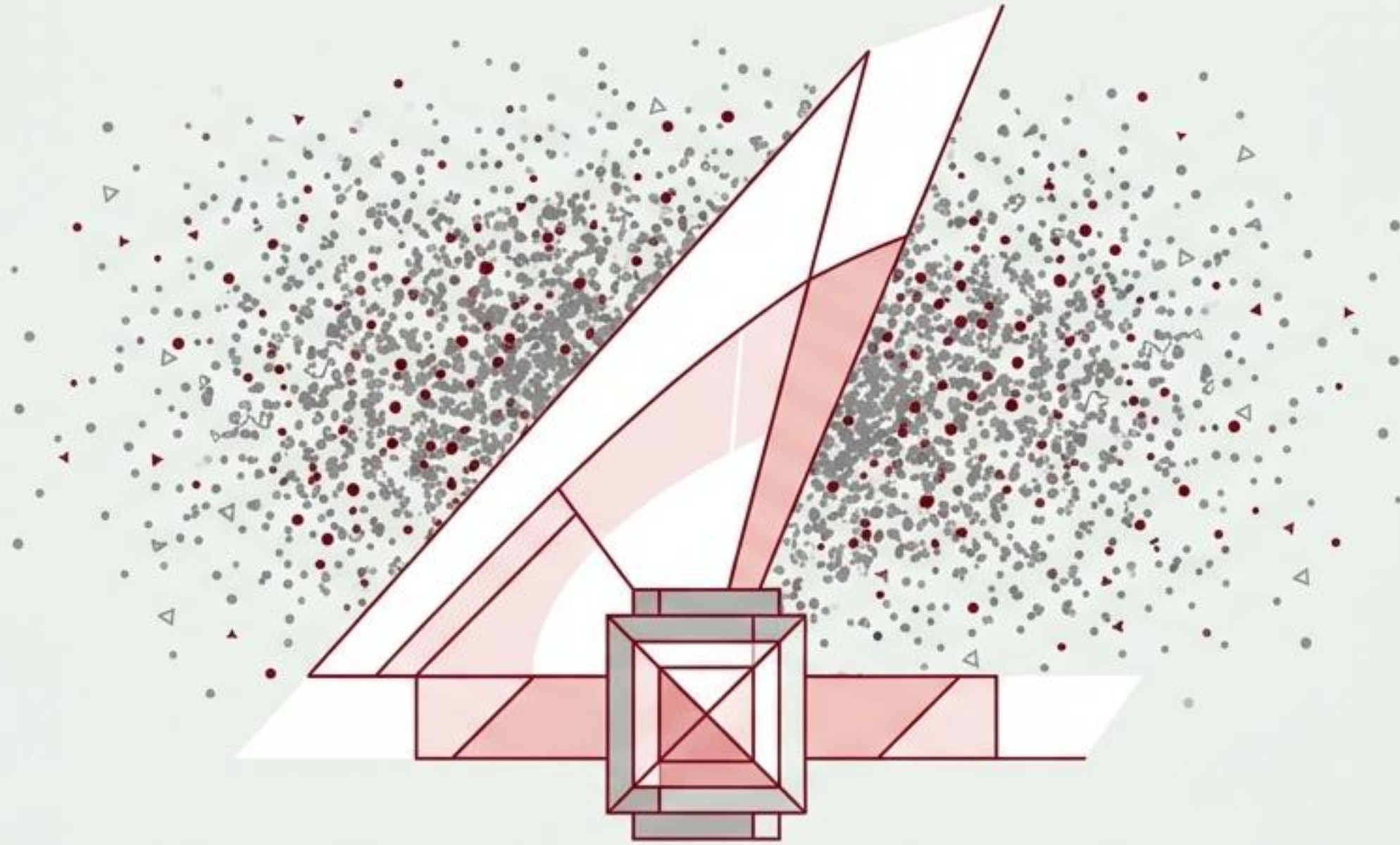
Overfitting (Data Snooping)



Finding a random pattern by testing 10 million variations. Strategy requires an Economic Premise, not just a pattern.

Warning: A perfect fit in-sample often fails out-of-sample (curve fitting).

The Statistical Lantern



Time Series Analysis is not a crystal ball.
It is a lantern that clarifies probabilities in a dark room.

The Process: Clean Noise → Transform Stationarity → Model Structure → Validate Rigorously.